



Day 9 Notes



Affine Transforms – A Shirt That Follows Me!

★ Today = the core of the whole project. A clothing image now follows my body live, using the pose points from Day 7.

1 What is an Affine Transform?

Moving, rotating and scaling an image so it lines up with reference points. My two shoulder points tell me three things:

- WHERE the shirt goes → midpoint between the shoulders
- HOW BIG it should be → distance between the shoulders
- HOW TILTED it should be → angle of the line between the shoulders

2 The Body Points I Used

11 = left shoulder	12 = right shoulder
23 = left hip	24 = right hip

Landmark values come back as 0 to 1, so I convert them to real pixels first:

```
function toPixels(p) {  
  return { x: p.x * canvas.width, y: p.y * canvas.height }  
}
```

3 The Three Measurements

```
// WHERE: midpoint of the shoulders  
const center = {  
  x: (leftShoulder.x + rightShoulder.x) / 2,  
  y: (leftShoulder.y + rightShoulder.y) / 2,  
}  
  
// HOW BIG: distances in pixels  
const shoulderWidth = Math.hypot(  
  leftShoulder.x - rightShoulder.x,  
  leftShoulder.y - rightShoulder.y)  
const torsoLength = Math.hypot(  
  center.x - hipCenter.x, center.y - hipCenter.y)  
  
// HOW TILTED: angle of the shoulder line
```

```
const angle = Math.atan2(
  leftShoulder.y - rightShoulder.y,
  leftShoulder.x - rightShoulder.x)
```

- `Math.hypot(dx, dy)` → distance between two points
- `Math.atan2(dy, dx)` → angle of a line, in radians

4 *Drawing the Shirt*

```
const shirtWidth = shoulderWidth * 2
const shirtHeight = torsoLength * 1.4

ctx.save()
ctx.translate(center.x, center.y)
ctx.rotate(angle)
ctx.drawImage(shirt, -shirtWidth / 2, -shirtHeight * 0.12,
              shirtWidth, shirtHeight)
ctx.restore()
```

- `translate` → moves the drawing origin to my shoulder midpoint
- `rotate` → tilts everything drawn after it
- `drawImage` → stamps the shirt PNG, centered on that point
- `save` / `restore` → undoes the move after each frame, so transforms don't pile up

The 2 and 1.4 multipliers are guesses (shirt wider than shoulders for sleeves, about as long as the torso). They need tuning later.

5 *What I Used*

A placeholder transparent shirt.png saved next to index.html. The same file will be replaced by real product images from the store later. I set segmentation aside today to keep things simple.



Big takeaway:

Pose points become measurements: midpoint = position, distance = size, angle = tilt. `translate` + `rotate` + `drawImage` turns those measurements into a shirt that follows me. Next (Day 10): make it look better, by smoothing the jitter, tuning the size, hiding the shirt when I'm out of frame, and trying a mirror view.